

Signals, Convolution & Z-Transforms

Revision notes with 10 fully solved questions

Scope: continuous-time signals, sampling and Nyquist rate, discrete-time indexing, convolution, zero-padding and introductory Z-transforms. Includes the material from *IDSA Assignment 3 pt1*, with 10 main solved questions (some multipart) across Sections 1–6.

Conventions: finite sequences are zero outside their listed indices. Unlabelled lists start at $n = 0$. In a stem plot, horizontal position is the sample index and vertical height is the sample value. Here $u[0] = 1$.

1 Read a signal before calculating

Continuous time: what does $x(t)$ mean?

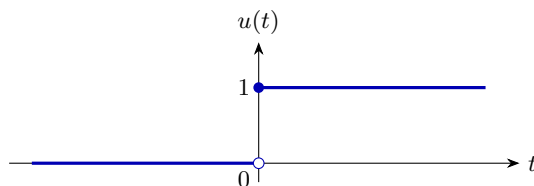
A **signal** describes how a quantity changes. For example, $x(t)$ could be a voltage measured at time t . A **continuous-time (CT) signal** has a value for every real time in its domain: $t = 0, 0.1, 0.12$, and every time between them. In a graph, the horizontal axis is time t and the vertical axis is the signal value (its **amplitude**).

Continuous time does not mean a smooth graph. A CT signal can jump. “Continuous-time” describes the time variable, not whether $x(t)$ is a continuous function. We use parentheses, $x(t)$, for CT signals and square brackets, $x[n]$, for discrete-time signals.

1. Unit step: a switch that turns on at zero.

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases} \quad u(t-a) \text{ turns on at } t = a.$$

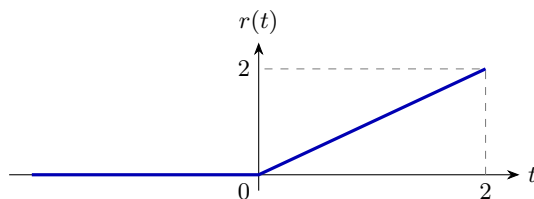
We choose $u(0) = 1$ in these notes. A filled dot marks an included endpoint; an open circle marks an excluded endpoint. No vertical connecting line is part of the step's graph.



2. Unit ramp: zero before the origin, then a straight rise.

$$r(t) = t u(t) = \begin{cases} 0, & t < 0, \\ t, & t \geq 0. \end{cases}$$

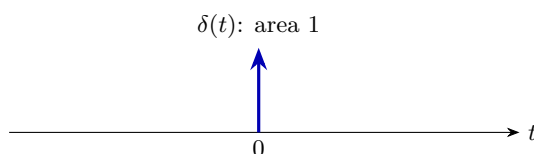
Its slope is 1: increasing time by one unit increases the amplitude by one unit.



3. Unit impulse: an ideal event with unit area. The CT impulse $\delta(t)$ is an idealized *distribution*, not an ordinary function with a finite peak. It is zero away from $t = 0$ and satisfies

$$\int_{-\infty}^{\infty} \delta(t) dt = 1, \quad \int_{-\infty}^{\infty} f(t) \delta(t-a) dt = f(a)$$

for f continuous at a . The arrow below is a symbol; its label gives **area (weight)**, not height. In contrast, $\delta[n]$ is an ordinary discrete sequence with value 1 at $n = 0$.

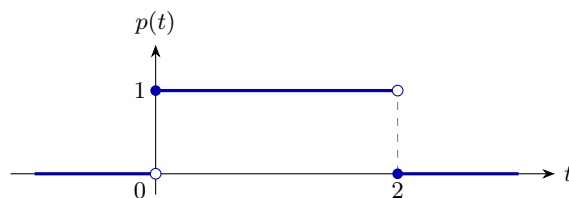


More continuous-time signals and their graphs

4. Rectangular pulse: switch on, then switch off. A pulse of height A from $t = a$ to $t = b$ ($a < b$) is

$$p(t) = A[u(t - a) - u(t - b)] = \begin{cases} A, & a \leq t < b, \\ 0, & \text{otherwise.} \end{cases}$$

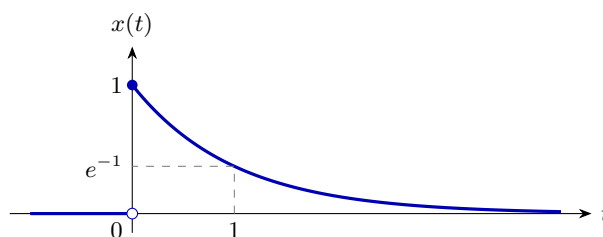
The first step turns it on; subtracting the second turns it off. Its width is $b - a$ and its area is $A(b - a)$. Below, $A = 1$, $a = 0$, $b = 2$.



5. Decaying exponential: a value that gradually dies away.

$$x(t) = e^{-at}u(t), \quad a > 0.$$

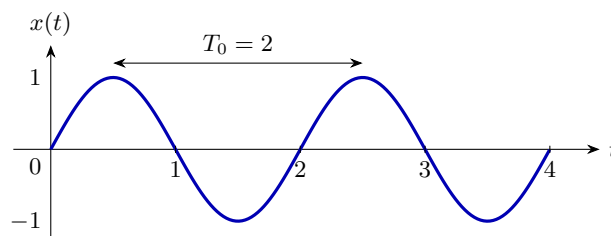
It is zero for $t < 0$, starts at 1 at $t = 0$, and approaches zero as $t \rightarrow \infty$. Larger a means faster decay. The **time constant** is $\tau = 1/a$; at $t = \tau$, the value is $e^{-1} \approx 0.368$. The graph uses $a = 1$.



6. Sinusoid: a repeating oscillation.

$$x(t) = A \sin(\omega_0 t + \phi), \quad \omega_0 = 2\pi f_0, \quad T_0 = \frac{1}{f_0} = \frac{2\pi}{\omega_0}.$$

For $A > 0$, A is the peak amplitude. With time measured in seconds, $f_0 > 0$ is frequency in hertz (cycles per second), ω_0 is angular frequency in radians per second, and ϕ is phase in radians (the starting position in the cycle). The **period** T_0 is the time for one complete cycle. The graph is $x(t) = \sin(\pi t)$, so $A = 1$, $\phi = 0$, $f_0 = 1/2$ and $T_0 = 2$.



A signal is **periodic** if $x(t + T) = x(t)$ for every t , for some $T > 0$. A sinusoid repeats for all time; only two cycles are displayed above. A single rectangular pulse and a decaying exponential are aperiodic.

How to shift, reflect and scale a continuous-time graph

Change the **outside** of $x(\cdot)$ to change heights. Change its **argument** to move points along the time axis. For $t_0 > 0$ and $a > 0$:

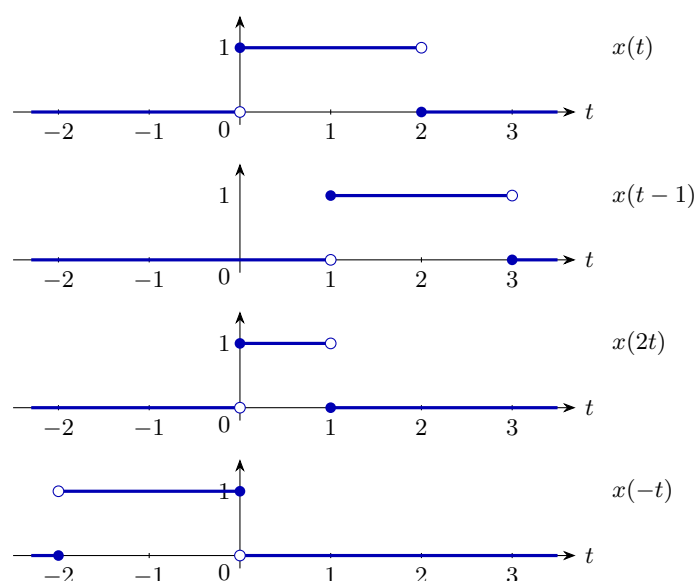
Expression	Effect on the original graph
$Cx(t)$	Multiply every height by C ; negative C also reflects vertically.
$x(t - t_0)$	Delay: move the graph right by t_0 .
$x(t + t_0)$	Advance: move the graph left by t_0 .
$x(-t)$	Time reversal: reflect across the vertical axis.
$x(at)$	Divide every time coordinate by a : compress if $a > 1$, stretch if $0 < a < 1$.

Reliable rule for a combined change: for $y(t) = Cx(at + b)$, with $a \neq 0$, a point originally at time s moves to

$$t = \frac{s - b}{a}, \quad \text{new height} = Cx(s).$$

This comes directly from setting the argument $at + b = s$. If $a < 0$, the order of the endpoints reverses. Apply this to the start, end, peak and every corner of a graph.

Graph illustration. Let $x(t) = u(t) - u(t - 2)$, the unit-height pulse on $0 \leq t < 2$. Each panel uses the same time scale; endpoint dots follow the chosen $u(0) = 1$ convention.



The nonzero intervals are $[0, 2)$, $[1, 3)$, $[0, 1)$ and $(-2, 0]$, respectively. Here $[]$ includes an endpoint and $()$ excludes it. Time scaling changes the width, not the height.

Connecting CT to discrete time. Sampling a CT signal every $T_s > 0$ time units gives $x[n] = x(nT_s)$, where n is an integer. A CT graph shows values at all times; a discrete-time stem plot shows values only at the allowed indices. Joining stems with a line does not define the missing values.

Sampling: converting a continuous-time signal into a sequence

Sampling means reading a signal at equally spaced times. If the spacing is T_s seconds, the sampling times are $\dots, -T_s, 0, T_s, 2T_s, \dots$ and

$$x[n] = x(t)|_{t=nT_s} = x(nT_s), \quad T_s = \frac{1}{f_s}.$$

Here n is an integer sample index, T_s is the **sampling interval** (seconds per sample), and f_s is the **sampling rate** (samples per second, conventionally expressed in Hz). Sampling chooses times; it does not round the amplitudes. Rounding amplitudes is a separate operation called quantization.

Three frequencies that must not be confused:

Symbol	Units	Meaning and conversion
f	Hz	Cycles per second in the original CT signal.
ω	rad/s	CT angular frequency: $\omega = 2\pi f$.
Ω	rad/sample	DT angular frequency: $\Omega = \omega T_s = 2\pi f / f_s$.

Thus the general sampling rule for a sinusoid is

$$A \cos(2\pi f t + \phi) \longrightarrow A \cos\left(2\pi \frac{f}{f_s} n + \phi\right).$$

The amplitude A and phase ϕ stay the same; replace *every* t by n/f_s .

Nyquist rate and why the highest frequency matters

A **band-limited** signal has no frequency components above some highest frequency f_m . Ideal uniform samples allow exact reconstruction when the sampling rate is greater than twice that highest frequency, with ideal reconstruction and no noise:

$$\text{Nyquist rate} = 2f_m, \quad f_s > 2f_m \text{ avoids the boundary case.}$$

Your assignment writes the threshold rule as $f_s \geq 2f_m$ and uses $f_s = 2f_m$ in its examples. We use those same rates below to reproduce the assignment's sequences. **Equality needs care:** a sinusoid exactly at $f_s/2$ can lose phase information. For example, $\sin(2\pi f_m n / (2f_m)) = \sin(\pi n) = 0$ at every sample. So equality is not a universal guarantee of perfect reconstruction.

The **Nyquist frequency** of a chosen sampler is $f_s/2$. This is different from the **Nyquist rate** $2f_m$ of a signal. **Aliasing** means different CT frequencies produce the same samples; undersampling can make a fast oscillation appear slower. An anti-aliasing low-pass filter removes unwanted high frequencies before sampling.

For a sum of sinusoids, find each $f = |\omega|/(2\pi)$ and take the largest. Amplitudes do not determine the Nyquist rate. A constant term is **DC**, meaning frequency zero.

1. Read the coefficient of t inside each sine or cosine; that is ω .
2. Divide by 2π to get Hz; choose the largest f_m .
3. Compute $2f_m$. Use the specified f_s , or state the rate you choose.
4. Put $T_s = 1/f_s$ and substitute $t = nT_s$ in the entire expression.
5. Use radians to calculate samples; plot each value at its integer index.

If only a Nyquist rate is requested, report $2f_m$ in Hz. If $x[n]$ is also requested but f_s is unspecified, its formula depends on the rate chosen; state that choice explicitly.

Question 1 — Sampling a single cosine: Assignment 3, sampling Q1

Given $x(t) = 2 \cos(10\pi t)$, find the Nyquist rate and the sampled sequence. Use the assignment's choice $f_s = 2f_m$ and plot the result.

Solution. Compare with $A \cos(2\pi f t + \phi)$:

$$A = 2, \quad \phi = 0, \quad 2\pi f_m = 10\pi \implies f_m = 5 \text{ Hz.}$$

The amplitude 2 does not affect this frequency. The CT period is $1/5 = 0.2$ s.

Nyquist rate = $2(5) = 10$ Hz.

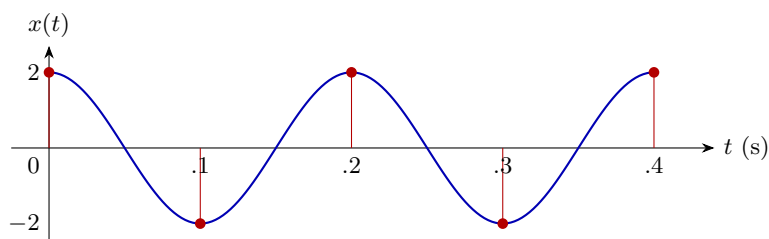
Choose $f_s = 10$ Hz as in the assignment. Then $T_s = 1/10 = 0.1$ s and

$$\begin{aligned} x[n] &= x(nT_s) = 2 \cos(10\pi n T_s) \\ &= 2 \cos\left(10\pi n \frac{1}{10}\right) = \boxed{2 \cos(\pi n) = 2(-1)^n}. \end{aligned}$$

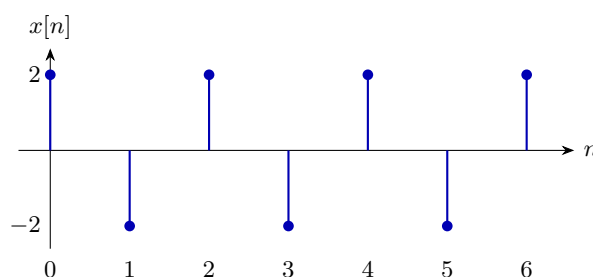
Even indices give 2 and odd indices give -2 :

n	Time nT_s (s)	Calculation	$x[n]$
0	0	$2 \cos 0$	2
1	0.1	$2 \cos \pi$	-2
2	0.2	$2 \cos 2\pi$	2
3	0.3	$2 \cos 3\pi$	-2
4	0.4	$2 \cos 4\pi$	2

CT curve and the sampling instants



Discrete sequence: $x[n] = 2(-1)^n$



Check: $x[n+2] = x[n]$, so the sequence has period 2 samples. It exists for all integer n ; only a short segment is drawn. At any other rate, use $x[n] = 2 \cos(10\pi n / f_s)$ instead.

Question 2 — Sampling a sum of cosines: Assignment 3, sampling Q2

Given

$$x(t) = 5 + 3 \cos(100\pi t) + 7 \cos(1200\pi t),$$

find the Nyquist rate and the sampled sequence, calculate sample values, and draw its stem plot using the assignment's threshold rate.

Solution. The constant 5 has frequency zero. Divide each angular frequency by 2π :

$$f_1 = \frac{100\pi}{2\pi} = 50 \text{ Hz}, \quad f_2 = \frac{1200\pi}{2\pi} = 600 \text{ Hz}.$$

Therefore $f_m = \max(0, 50, 600) = 600 \text{ Hz}$ and

$$\boxed{\text{Nyquist rate} = 2(600) = 1200 \text{ Hz.}}$$

Choose $f_s = 1200 \text{ Hz}$, giving $T_s = 1/1200 \text{ s} \approx 0.8333 \text{ ms}$. Substitute $t = n/1200$ into both cosine terms:

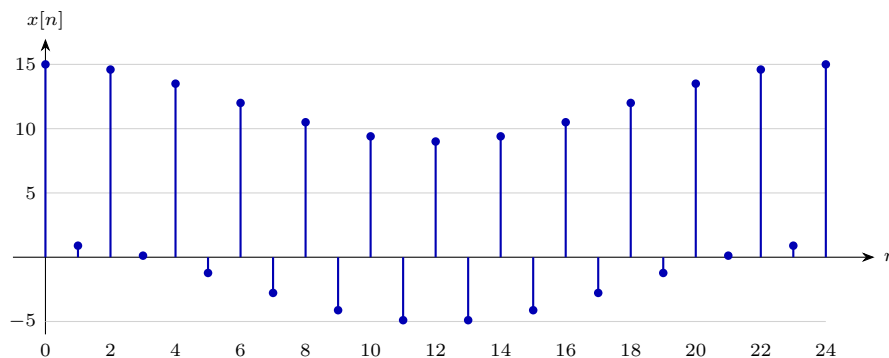
$$\begin{aligned} x[n] &= 5 + 3 \cos\left(\frac{100\pi n}{1200}\right) + 7 \cos\left(\frac{1200\pi n}{1200}\right) \\ &= \boxed{5 + 3 \cos\left(\frac{\pi n}{12}\right) + 7 \cos(\pi n)} \\ &= 5 + 3 \cos\left(\frac{\pi n}{12}\right) + 7(-1)^n. \end{aligned}$$

The two DT angular frequencies are $\pi/12$ and π radians/sample. The DC value remains 5.

n	Calculation	$x[n]$ (approximately)
-1	$5 + 3 \cos(-\pi/12) - 7$	0.89778
0	$5 + 3(1) + 7$	15
1	$5 + 3 \cos(\pi/12) - 7$	0.89778
2	$5 + 3 \cos(\pi/6) + 7$	14.59808
3	$5 + 3 \cos(\pi/4) - 7$	0.12132
4	$5 + 3 \cos(\pi/3) + 7$	13.5

For instance, $x[1] = 5 + 3(0.9659258) - 7 = 0.8977774$. Use radian mode when entering $\pi/12$ into a calculator. Since cosine is even, $x[-n] = x[n]$.

One full period, with the repeated endpoint at $n = 24$



Check: $\cos(\pi n/12)$ repeats every 24 samples and $\cos(\pi n)$ every 2 samples. The combined sequence has period $\text{lcm}(24, 2) = 24$, so $x[24] = x[0] = 15$. As in Question 1, the rate used is exactly the textbook threshold; choose $f_s > 1200 \text{ Hz}$ to avoid its boundary ambiguity.

Discrete-time indexing

A discrete-time signal $x[n]$ assigns a value to each integer index n . An arrow under a sequence entry marks $n = 0$. Always write the index row before calculating.

- $\delta[n - r]$ is 1 at $n = r$ and zero elsewhere.
- $u[n]$ is 1 for $n \geq 0$ and zero for $n < 0$.
- $u[n] - u[n - L]$ is a length- L rectangle: 1 at $n = 0, \dots, L - 1$.
- For $v[n] = x[an + b]$, a sample originally at m appears where $an + b = m$, if that gives an integer n ; its height stays unchanged.

Question 3 — Indexing, shifting and reflection

Given $x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}$, find $x[-1]$, $x[3]$, $x[n + 1]$ and $x[1 - n]$. Also express $x[n]$ as shifted impulses.

Solution. The original samples are

$$\begin{array}{c|cccc} n & -2 & -1 & 0 & 1 \\ \hline x[n] & 1 & 2 & 2 & 1 \end{array}.$$

Thus $\boxed{x[-1] = 2, \quad x[3] = 0}$, since 3 lies outside the listed range.

For $v[n] = x[n + 1]$, solve $n + 1 = m$, giving $n = m - 1$. For $w[n] = x[1 - n]$, solve $1 - n = m$, giving $n = 1 - m$:

Original index m	Height $x[m]$	Index in $v[n]$	Index in $w[n]$
-2	1	-3	3
-1	2	-2	2
0	2	-1	1
1	1	0	0

Reading each result in increasing index order,

$$\boxed{x[n + 1] = \{1, 2, 2, 1\}_{n=-3, -2, -1, 0}}, \quad \boxed{x[1 - n] = \{1, 2, 2, 1\}_{n=0, 1, 2, 3}}.$$

Place an impulse at each original index and multiply it by that sample's height:

$$\boxed{x[n] = \delta[n + 2] + 2\delta[n + 1] + 2\delta[n] + \delta[n - 1]}.$$

Check: substituting $n = -2, -1, 0, 1$ gives 1, 2, 2, 1, respectively. Every other index gives zero.

2 Linear convolution: flip, shift, multiply, add

For a linear time-invariant system with impulse response $h[n]$, each input sample produces a scaled, shifted copy of $h[n]$. Adding these copies gives

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

Here n is the output index and k is the summation index. Convolution is a sum of overlap products, not pointwise multiplication.

1. **Index:** if the input ranges are $a, \dots, b$ and $c, \dots, d$, the output lies within $a + c, \dots, b + d$.
2. **Flip:** change $h[k]$ to $h[-k]$.
3. **Shift:** form $h[n - k]$; a sample of h originally at m is now at $k = n - m$.
4. **Multiply and add:** multiply at matching k and sum. Repeat for each n .

For listed lengths L, M , retain $L + M - 1$ output positions, including any zero entries.

Question 4 — Linear convolution with negative indices

Find $y = x * h$ for

$$x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}, \quad h = \{1, 3, 2\}_{n=-1, 0, 1}.$$

Solution. The output range is $-2 + (-1) = -3$ to $1 + 1 = 2$, with $4 + 3 - 1 = 6$ positions. The flipped sequence $h[-k]$ has heights 2, 3, 1 at $k = -1, 0, 1$.

Using the four possible nonzero samples of x ,

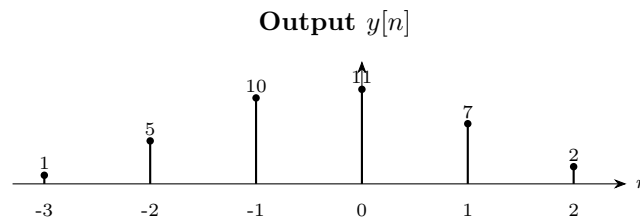
$$y[n] = h[n+2] + 2h[n+1] + 2h[n] + h[n-1].$$

Substitute each output index; only nonzero products are shown:

n	Overlap products	$y[n]$
-3	1(1)	1
-2	1(3) + 2(1)	5
-1	1(2) + 2(3) + 2(1)	10
0	2(2) + 2(3) + 1(1)	11
1	2(2) + 1(3)	7
2	1(2)	2

For instance, at $n = -2$, the shifted sequence $h[-2 - k]$ has heights 2, 3, 1 at $k = -3, -2, -1$. Only $k = -2, -1$ overlap x , so $y[-2] = 1(3) + 2(1) = 5$.

$$y = \{1, 5, 10, 11, 7, 2\}_{n=-3, -2, -1, 0, 1, 2}.$$



Check: $\sum_n y[n] = 1 + 5 + 10 + 11 + 7 + 2 = 36 = (1 + 2 + 2 + 1)(1 + 3 + 2)$. Outside $[-3, 2]$, there is no overlap and $y[n] = 0$.

3 Overlap counting and shifted rows

For two rectangles of unit height, every nonzero product is 1, so convolution counts overlapping samples. For general sequences, a shifted-row table organizes the same multiply–add calculation.

Question 5 — Overlap counting and signed shifted rows

(a) Convolution of two rectangular sequences. Find $y = x * h$ when $x = h = \{1, 1, 1\}$, both starting at $n = 0$.

Solution. Both inputs occupy 0, 1, 2, so y occupies 0, $\dots$, 4. The fixed sequence $x[k]$ is nonzero for $0 \leq k \leq 2$. The moving sequence $h[n - k]$ is nonzero when $0 \leq n - k \leq 2$, or $n - 2 \leq k \leq n$.

n	Nonzero k for $h[n - k]$	Overlapping k	Sum	$y[n]$
0	$-2, -1, 0$	$\{0\}$	1	1
1	$-1, 0, 1$	$\{0, 1\}$	$1 + 1$	2
2	$0, 1, 2$	$\{0, 1, 2\}$	$1 + 1 + 1$	3
3	$1, 2, 3$	$\{1, 2\}$	$1 + 1$	2
4	$2, 3, 4$	$\{2\}$	1	1

$$y = \{1, 2, 3, 2, 1\}_{n=0,1,2,3,4}.$$

Check: there are $3 + 3 - 1 = 5$ output positions and $1 + 2 + 3 + 2 + 1 = 9 = 3 \cdot 3$.

(b) Signed samples using shifted rows. Find $\{2, -1, 1\} * \{1, 2\}$, with both lists starting at zero.

Solution. The output has $3 + 2 - 1 = 4$ positions. Since $x[0] = 2$, $x[1] = -1$ and $x[2] = 1$,

$$y[n] = 2h[n] - h[n - 1] + h[n - 2].$$

Write each scaled copy of h in its correct columns, then add:

Output index n	0	1	2	3
$2h[n]$	2	4	0	0
$-h[n - 1]$	0	-1	-2	0
$h[n - 2]$	0	0	1	2
$y[n]$	2	3	-1	2

Thus $y[0] = 2$, $y[1] = 4 - 1 = 3$, $y[2] = -2 + 1 = -1$ and $y[3] = 2$, giving

$$y = \{2, 3, -1, 2\}_{n=0,1,2,3}.$$

Check: $2 + 3 - 1 + 2 = 6 = (2 - 1 + 1)(1 + 2)$.

Useful properties and checks

$$x * h = h * x, \quad x * (h_1 + h_2) = x * h_1 + x * h_2, \quad x * \delta[n - r] = x[n - r].$$

Check the output length, the first and last products, and $\sum y = (\sum x)(\sum h)$. A correct total alone does not verify the order of the samples. Keep interior zeros and track the origin explicitly.

4 Circular convolution: rotating circles

An N -point circular convolution wraps all indices modulo N :

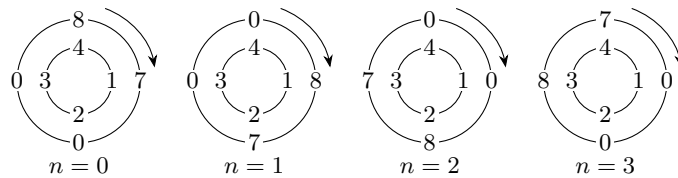
$$y_N[n] = \sum_{k=0}^{N-1} x[k]h[(n-k) \bmod N], \quad 0 \leq n < N.$$

Modulo N returns an index in $0, \dots, N-1$; for example, $(-1) \bmod 4 = 3$. The stored answer contains N samples; its periodic extension repeats these samples.

Question 6 — Four-point circular convolution with the ring method

Find the 4-point circular convolution of $x = \{1, 2, 3, 4\}$ and $h = \{7, 8\}$.

Solution. Pad the second input to $h = \{7, 8, 0, 0\}$. Place x clockwise on the inner ring, starting at the right. Place h anticlockwise on the outer ring. Read aligned products and add; then rotate the outer ring one position clockwise for the next output.



At fixed $k = 0, 1, 2, 3$ (right, down, left, up), the outer values and dot products are:

n	$(h[n], h[n-1], h[n-2], h[n-3])$	Dot product with $(1, 2, 3, 4)$
0	$(7, 0, 0, 8)$	$1(7) + 2(0) + 3(0) + 4(8) = 39$
1	$(8, 7, 0, 0)$	$1(8) + 2(7) + 3(0) + 4(0) = 22$
2	$(0, 8, 7, 0)$	$1(0) + 2(8) + 3(7) + 4(0) = 37$
3	$(0, 0, 8, 7)$	$1(0) + 2(0) + 3(8) + 4(7) = 52$

All h indices above are modulo 4. Each row is a right rotation of the preceding row.

$$y_4 = \{39, 22, 37, 52\}_{n=0,1,2,3}.$$

Check: $39+22+37+52 = 150 = (1+2+3+4)(7+8)$. The same result follows from $y_4[n] = 7x[n] + 8x[(n-1) \bmod 4]$.

Question 7 — Three-point circular convolution with a negative sample

Find the 3-point circular convolution of $x = \{1, 2, 0\}$ and $h = \{1, -1, 0\}$.

Solution. Circularly reverse h to get $(h[0], h[2], h[1]) = (1, 0, -1)$, then rotate right for each successive output:

n	Rotated h row	Dot product with $(1, 2, 0)$
0	$(1, 0, -1)$	$1(1) + 2(0) + 0(-1) = 1$
1	$(-1, 1, 0)$	$1(-1) + 2(1) + 0(0) = 1$
2	$(0, -1, 1)$	$1(0) + 2(-1) + 0(1) = -2$

$$y_3 = \{1, 1, -2\}_{n=0,1,2}.$$

Check: $1 + 1 - 2 = 0 = (1 + 2 + 0)(1 - 1 + 0)$.

5 Linear versus circular: choose the right length

	Linear convolution	N -point circular convolution
Index behaviour	Finite inputs are zero outside their ranges.	Input indices wrap modulo N .
Output length	$L + M - 1$ listed positions.	Exactly N stored positions.
Method	Flip, shift, multiply, add.	Reverse on a ring, rotate, multiply, add.
Tail	Remains at its own indices.	Folds onto indices with the same remainder modulo N .

Use the stated N and pad finite inputs with zeros as needed. If no N is given, $N = \max(L, M)$ is a common convention; state it explicitly. To reproduce the full linear result by circular convolution, the standard sufficient padding rule is

$$N \geq L + M - 1.$$

Repeating a sequence changes its samples; it is not zero-padding. An LCM may give a common period for periodic sequences, but is not the general padding rule for finite-list convolution.

Question 8 — Linear output, circular wrapping and zero-padding

(a) Calculate and compare the outputs. For $x = \{1, 2, 3, 4\}$ and $h = \{7, 8\}$, find the linear convolution and obtain the 4-point and 5-point circular results from it.

Solution. The linear output has $4 + 2 - 1 = 5$ positions. Calculate every sample:

$$\begin{aligned} y_{\text{lin}}[0] &= 1(7) = 7, \\ y_{\text{lin}}[1] &= 1(8) + 2(7) = 22, \\ y_{\text{lin}}[2] &= 2(8) + 3(7) = 37, \\ y_{\text{lin}}[3] &= 3(8) + 4(7) = 52, \\ y_{\text{lin}}[4] &= 4(8) = 32. \end{aligned}$$

Hence $y_{\text{lin}} = \{7, 22, 37, 52, 32\}$ at $n = 0, \dots, 4$. For $N = 4$, index 4 wraps to index 0, so

$$y_4[0] = 7 + 32 = 39, \quad y_4 = \{39, 22, 37, 52\}.$$

For $N = 5$, pad both inputs to length 5. No nonzero tail wraps into the start, giving

$$y_5 = \{7, 22, 37, 52, 32\} = y_{\text{lin}}.$$

Check: both totals are $150 = (1 + 2 + 3 + 4)(7 + 8)$. The minimum length preserving all five linear positions is $N = 5$.

(b) Selecting a length from the input sizes. Two finite lists have lengths $L = 4$ and $M = 6$. Find $\max(L, M)$, $\text{lcm}(L, M)$ and the minimum length guaranteed to reproduce their full linear convolution using circular convolution.

Solution. The larger input length is $\max(4, 6) = 6$. Since $4 = 2^2$ and $6 = 2 \cdot 3$, their least common multiple is $2^2 \cdot 3 = 12$. The linear output length is

$$N_{\text{min}} = L + M - 1 = 4 + 6 - 1 = 9.$$

At $N = 6$, possible linear samples at indices 6, 7, 8 fold onto 0, 1, 2, respectively. At $N = 9$, append five zeros to the length-4 input and three zeros to the length-6 input. All nine output positions remain distinct. At $N = 12$, the linear output is also reproduced, followed by three zeros, but the extra padding is unnecessary.

6 Z-transform: attach a power to each sample

The introductory Z-transform represents each sample as a coefficient and its index as a power:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

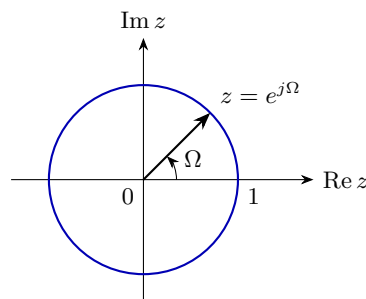
Write the indices first, form $x[n]z^{-n}$ for every nonzero sample, and add. Negative indices give positive powers of z . For an infinite series, state its convergence condition. Only this basic convergence condition is used here.

What is z ? Reading the circle in the assignment

The transform variable z is a complex number. In polar form,

$$z = re^{j\Omega} = r(\cos \Omega + j \sin \Omega), \quad j^2 = -1.$$

Here $r = |z|$ is the distance from the origin and Ω is the angle in radians. The **unit circle** has radius 1, so $z = e^{j\Omega}$ on that circle. The sketch describes the transform variable, not a time-domain signal.



For each sample, the exponent is *minus its index*: $n = -1$ gives z^1 , $n = 0$ gives $z^0 = 1$, and $n = 1$ gives $z^{-1} = 1/z$. The **region of convergence** (ROC) is the set of z values where the transform converges. The finite sequences below need only exclude $z = 0$ (among finite complex values), because their expressions contain negative powers.

Question 9 — Finite-sequence Z-transforms and convolution

(a) Assignment Z-transform Q1: $x = \{2, 4, 7\}_{n=0,1,2}$. Find $X(z)$ directly from the definition.

Solution. The first value is at $n = 0$, so

n	0	1	2
$x[n]$	2	4	7
$x[n]z^{-n}$	$2z^0$	$4z^{-1}$	$7z^{-2}$

Adding all three terms,

$$\begin{aligned}
 X(z) &= \sum_{n=0}^2 x[n]z^{-n} = 2 + 4z^{-1} + 7z^{-2} \\
 &= 2 + \frac{4}{z} + \frac{7}{z^2} = \boxed{\frac{2z^2 + 4z + 7}{z^2}}, \quad z \neq 0.
 \end{aligned}$$

Assignment sequence (a)



Check: $X(1) = 2 + 4 + 7 = 13$, equal to the sum of the samples.

(b) Assignment Z-transform Q2: $x = \{3, 4, 7\}_{n=-1,0,1}$. The origin arrow is under the middle value 4. Find $X(z)$.

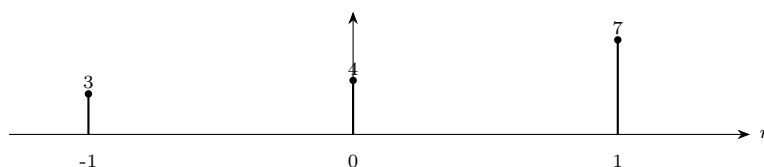
Solution. The indices are now $-1, 0, 1$, not $0, 1, 2$:

n	-1	0	1
$x[n]$	3	4	7
$x[n]z^{-n}$	$3z$	4	$7z^{-1}$

Therefore

$$\begin{aligned}
 X(z) &= \sum_{n=-1}^1 x[n]z^{-n} = 3z + 4 + 7z^{-1} \\
 &= 3z + 4 + \frac{7}{z} = \boxed{\frac{3z^2 + 4z + 7}{z}}, \quad z \neq 0.
 \end{aligned}$$

Assignment sequence (b)



Check: $X(1) = 3 + 4 + 7 = 14$. The sample at negative index produces a *positive* power of z ; moving the origin changes the transform.

Question 9 continued: (c) Convolution through Z-transforms

Find $X(z)$ and $H(z)$ for

$$x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}, \quad h = \{1, 3, 2\}_{n=-1, 0, 1},$$

and multiply them to obtain the transform of their linear convolution.

Solution. Attach z^{-n} to each sample:

Index n	-2	-1	0	1
$x[n]$	1	2	2	1
$x[n]z^{-n}$	z^2	$2z$	2	z^{-1}

Therefore

$$\boxed{X(z) = z^2 + 2z + 2 + z^{-1}}, \quad \boxed{H(z) = z + 3 + 2z^{-1}}.$$

When multiplying transforms, exponents add. The coefficient at z^{-n} is $\sum_k x[k]h[n-k]$, so convolution corresponds to multiplication: $Y(z) = X(z)H(z)$.

$$\begin{aligned} Y(z) &= z^2(z + 3 + 2z^{-1}) + 2z(z + 3 + 2z^{-1}) \\ &\quad + 2(z + 3 + 2z^{-1}) + z^{-1}(z + 3 + 2z^{-1}) \\ &= (z^3 + 3z^2 + 2z) + (2z^2 + 6z + 4) \\ &\quad + (2z + 6 + 4z^{-1}) + (1 + 3z^{-1} + 2z^{-2}) \\ &= \boxed{z^3 + 5z^2 + 10z + 11 + 7z^{-1} + 2z^{-2}}. \end{aligned}$$

The coefficients at indices $-3, -2, -1, 0, 1, 2$ give $\{1, 5, 10, 11, 7, 2\}$, agreeing with Question 4. Each expression is finite for finite $z \neq 0$; $z = 0$ is excluded because negative powers occur.

Question 10 — Transform of a right-sided exponential

Find the Z-transform of $x[n] = (1/3)^n u[n]$ directly from the definition.

Solution. Since $u[n] = 0$ for $n < 0$ and $u[n] = 1$ for $n \geq 0$,

$$X(z) = \sum_{n=0}^{\infty} (1/3)^n z^{-n} = 1 + \frac{1}{3}z^{-1} + \frac{1}{9}z^{-2} + \frac{1}{27}z^{-3} + \dots$$

This is a geometric series with ratio $r = \frac{1}{3}z^{-1}$. Use $\sum_{n=0}^{\infty} r^n = 1/(1-r)$ when $|r| < 1$:

$$\boxed{X(z) = \frac{1}{1 - \frac{1}{3}z^{-1}} = \frac{z}{z - \frac{1}{3}}, \quad |z| > \frac{1}{3}}.$$

Check: the coefficients of $1, z^{-1}, z^{-2}, z^{-3}$ are $1, 1/3, 1/9, 1/27$, exactly the first four samples. More generally, $a^n u[n]$ has transform $1/(1 - az^{-1})$ for $|z| > |a|$, with $a \neq 0$.